

Forest Fires Classification

CSCU9M5 Assignment 1
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Methodology (CRISP-DM)

Import forestfires_classification.csv into Orange, check and understand data features, find target variable.

Select row to remove FFMC outlier, select column to keep only necessary variables as predictor.

Utilize preprocess widget to normalize (z-score) to all numeric predictors.

Utilize data sampler to split the data once into 80% for training & 20% for testing.

Compared Logistic Regression, SVM, Random Forest using Test & Score Widget with 10-fold cross validation parameter.

Select the best model (Random Forest: 300 trees), retrain it on the full 80% and test it on the 20% test set.

Record final testing results and number of instances from confusion matrix, evaluate and interpret the data.

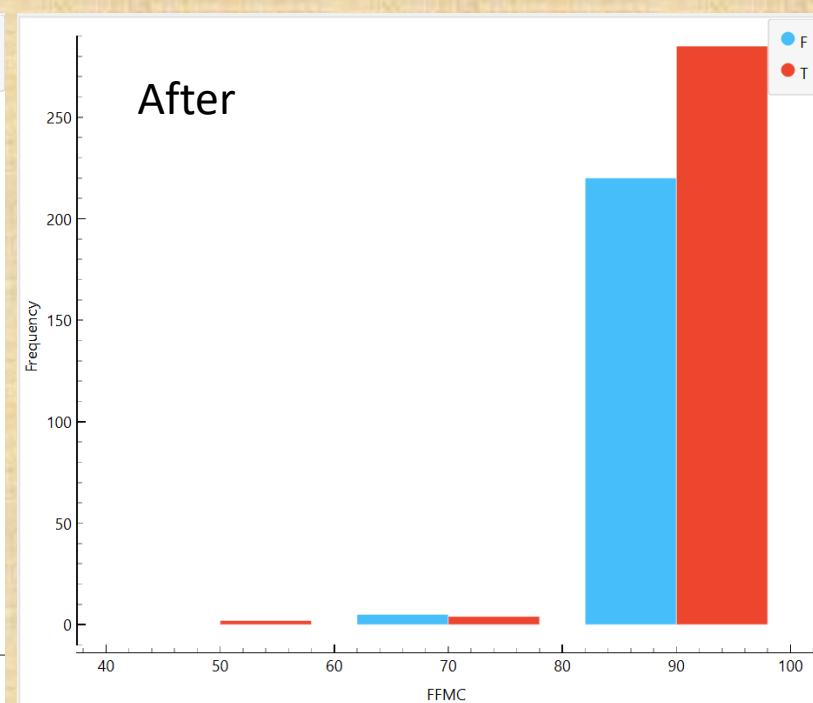
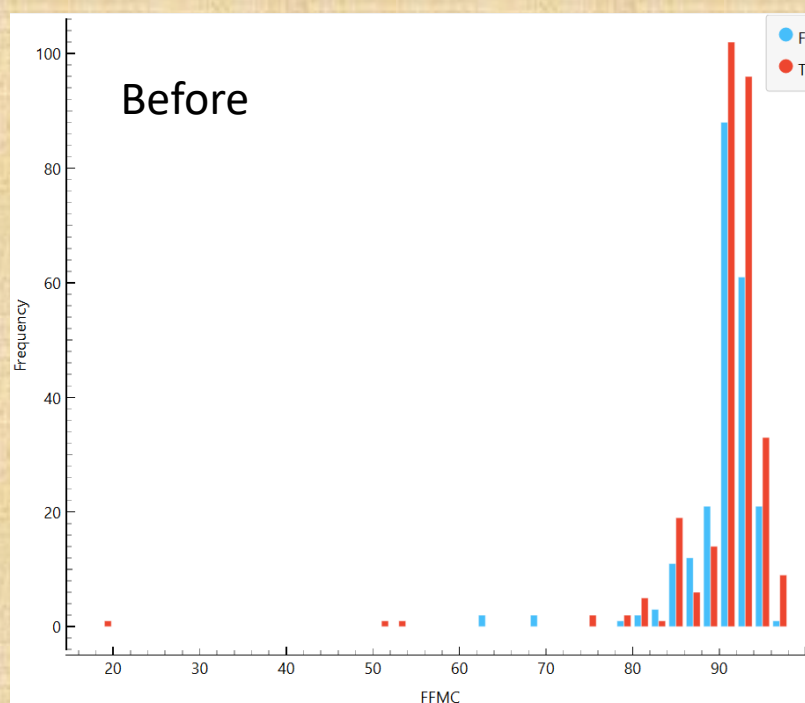
Data Understanding

The model uses the fire-danger indices and weather variables, because these variables directly describe physical conditions that drive fire severity (Cortez and Morais, 2007). These predictors were treated as numeric continuous features and standardized with z-scores

Variables	Type	Description & Role
FFMC	Numeric continuous	Fine Fuel Moisture Code (Predictor)
DMC	Numeric continuous	Duff Moisture Codde (Predictor)
DC	Numeric continuous	Drought Code (Predictor)
ISI	Numeric continuous	Initial Spread Index (Predictor)
temp	Numeric continuous	Air Temperature (C) (Predictor)
RH	Numeric continuous	Relative Humidity (%) (Predictor)
wind	Numeric continuous	Wind Speed (km/h) (Predictor)
rain	Numeric continuous	Rainfall (mm) (Predictor)
area	Categorical Binary	Burned area (Target)

Data Preparation

Variable FFMC has outlier and was removed using the select row widget. FFMC was set with > 30 condition. Removing the 18.7 extreme low value.



Modelling

Model	Best Parameters	Best Validation Metrics
Logistic Regression	C = 10,	AUC = 0.650, CA = 0.578, F1 = 0.578, Prec = 0.577, Recall = 0.581.
SVM	C = 10, g = auto	AUC = 0.759, CA = 0.692, F1 = 0.687, Prec = 0.689, Recall = 0.692.
Random Forest	300, default depth	AUC = 0.939, CA = 0.872, F1 = 0.872, Prec = 0.874, Recall = 0.872.
Final Test 20% of Data		
Random Forest	300, default depth	AUC = 0.885, CA = 0.767, F1 = 0.767, Prec = 0.789, Recall = 0.767.

For Logistic Regression and SVM, C =0.1 ,1, 10 were tested; for Random Forest, 100, 200 and 300 trees were compared. The best hyperparameter of each was recorded. All models were evaluated with 10-fold cross-validation on the 80% training split using F1 (balance of precision/recall) and AUC (area under the curve class separation). The best model Random Forest with F1 = 0.872 and AUC = 0.939 was selected as it captures non-linear variables the best (Gupta, 2021) . On the 20% test set, the Random Forest achieved 76.7% acceptable accuracy (F1 = 0.767, AUC = 0.885).

Results & Error

Actual/Predicted	F	T	Σ
F	39	6	45
T	18	40	58
Σ	57	46	103

On the 20% hold-out data, the model correctly classified 39 low-burn and 40 high-burn fires (79/103 cases). It made 18 false negatives and 6 false positives, so it tends to avoid raising alarm to minimize error, leading to unpredicted high burned fires.

Insights & Deployment

Rank widget in Orange shows that DC, DMC, FFMC and ISI, followed by temperature and humidity, are the main drivers of large burned areas, while wind and rain are less influential. The confusion matrix shows the model leans towards predicting low-burn, so it is cautious about calling a fire high-burn. This model is acceptable; it is accurate enough to be used as a supporting tools in wildfire management.

References

Chapman, P. et al. (2000) *CRISP-DM 1.0: Step-by-Step Data Mining Guide*. CRISP-DM Consortium / SPSS
Gupta, A. (2021) 'Understanding Random Forest', *Analytics Vidhya*, 21 June.
Cortez, P. & Morais, A. (2007) 'A data mining approach to predict forest fires using meteorological data', *Proc. EPIA 2007*.

